Effects of Social Media Use on Climate Change Opinion, Knowledge, and Behavior

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Summary

Early research on the relationship between social media use and its relationship to climate change opinion, knowledge, and behavior suggests several positive impacts. Social media encourages greater knowledge of climate change, mobilization of climate change activists, space for discussing the issue with others, and online discussions that frame climate change as a negative for society. Social media, however, does provide space for framing climate change skeptically and activating those with a skeptical perspective of climate change. Further examination of the relationship between social media use and climate change perceptions is warranted.

Keywords: social media use, climate change, public opinion, knowledge, behavior

Subjects: Communication

Social media use is an important area within climate change communication; however, research on the topic is quite nascent. Major perspectives on social media are discussed and existing research is outlined on different areas of climate change perceptions, including opinion, knowledge, and behavior, identifying how research on communication in social media forms has been applied to each area.

Social Media

An early and popular definition of social media states that it is an online structure where individuals use their own profiles to connect with other individuals by creating lists of friends' profiles (Boyd & Ellison, 2007). Expanding upon this, social media is one form of user-generated content, or online mediated content that is created outside of professionally produced and edited circles (Organisation for Economic Co-operation and Development, 2007). Such content ranges from posts on social networking sites as defined by Boyd and Ellison (2007) to other social content, such as comments people leave on news sites, reviews individuals write about restaurants, blog posts, and most-emailed articles lists on news websites. Here social media is conceptualized as *social* content online. This goes beyond the strict sense of the term *social networking sites* as defined by Boyd and Ellison (2007) to include other types of social user-generated content (i.e., non-professionally produced content). Social content, however, can include content produced by professionals (e.g., businesses and news organizations relying on social media platforms to communicate with audiences).

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Usage of different forms of social communication online is growing. For instance, between 2005 and 2013, the percentage of adults online in the United States who claimed to use social networking sites increased from 8% to 72% (Brenner & Smith, 2013). In addition, people are engaging in other social activities online. For instance, the majority of Americans (55%) have commented online, with most of those comments occurring on social media (Stroud, Van Duyn, & Peacock, n.d.). More than three-quarters have read a comment online, with the majority of those individuals reading comments a few times a month or less (Stroud et al., n.d.). These newer forms of social content are grounded in the usage of original forms of social content, like blogs, which 24% of adults in the United States read and almost 1 in 10 (9%) write or contribute to, according to a 2008 survey (Smith, 2008).

Scholars categorize social media use as both expressive and consumptive (e.g., Gil de Zuniga, Bachmann, Hsu, & Brundidge, 2013). Social media use as consumptive is thought of as a more passive use of the medium, with individuals seeking out news and information on social media platforms (Gil de Zúñiga, Jung, & Valenzuela, 2012) and news exposure on the most popular social media sites like Facebook being incidental, or something people come across while being on the site for other reasons (Gottfried & Shearer, 2016). This exposure is influenced by the fact that reading news on social media means it is filtered by friends and family and resides within the context of various tones that would not typically be present in a professional news setting. Scholars of computer-mediated communication have begun to explore how the social context of participatory media online changes how people interpret media (Lee, 2012; Metzger, Flanagin, & Medders, 2010; Walther & Jang, 2012) and issues in the media (Houston, Hansen, & Nisbett, 2011; Hsueh, Yogeeswaran, & Malinen, 2015).

Online social activities are increasingly important for how people consume news and information about a variety of important social issues. Most adults in the United States (62%) get news on social media (Gottfried & Shearer, 2016). Recent research examines news consumption on a variety of specific social media sites, including Reddit, a forum that allows individuals to create discussion posts and respond to and vote on others' posts to bring them to the top of the forum, and Facebook and Twitter, both social networking sites in which users create profiles, follow other profiles, and produce and reply to text, photo, or video status updates. The majority of those in the United States who use Reddit, Facebook, and Twitter get news while using the site, and 44% of all U.S. adults get news on Facebook (Gottfried & Shearer, 2016). People in the United States are commenting on online news stories and blog items (25% of Internet users) and sharing links to news on social media sites (17% of Internet users) (Purcell, Rainie, Mitchell, Rosenstiel, & Olmstead, 2010). This relationship between online social media and news use is important because it exemplifies the connection between how people use social media and how they perceive important social issues, including scientific issues such as climate change. For instance, individuals' perceptions about the news and issues in the news are associated with the comments to which the news stories are connected (Anderson, Brossard, Scheufele, Xenos, & Ladwig, 2014; Anderson, Yeo, Brossard, Scheufele, & Xenos, 2016; Houston et al., 2011; Lee, 2012), blog readership (Thorson, Vraga, & Ekdale, 2010), and social user–generated content (Metzger et al., 2010). In short, evidence shows that social media shapes public perceptions for a range of issues.

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Social media use is also expressive, meaning that it is about both opinion expression, or sharing or talking with others about preferences over social media, and about participation or activism (Boulianne, 2015; Valenzuela, 2013; Vraga, Anderson, Kotcher, & Maibach, 2015). "Expressive" blog use (e.g., posting comments) is related to political participation and deeper cognitive reasoning (Gil de Zuniga et al., 2013). Online political messaging and political talk is related to political participation (Gil de Zúñiga, Veenstra, Vraga, & Shah, 2010). Online interactions with others in social messaging services and on social media can encourage political participation (Rainie, Smith, Schlozman, Brady, & Verba, 2012; Shah et al., 2007). Research on the relationship between social media and political participation is grounded in other—typically earlier—research that shows a positive relationship between general Internet use not necessarily in the form of social communication and political participation (Scheufele & Nisbet, 2002; Xenos & Moy, 2007). Research has started to add dimensions to studying this topic, showing that certain characteristics of individuals such as political interest or conflict avoidance are related to being more likely to express an opinion on social media like Facebook (Vraga, Thorson, Kligler-Vilenchik, & Gee, 2015). Science communication scholars have pointed to the role social media can play in engaging wider publics in review and critique of scientific research after it has already been published (Yeo et al., 2016) and in encouraging informal interpersonal interactions among scientists alongside formal components of the scientific process, such as publication in peer-reviewed journals (Lievrouw, 2010). In addition, Internet platforms may provide more transparency to the scientific process by providing open access to publications and research data, often known as the "open science movement" (Lin, 2012). In short, one general perspective within the literature on social media is that expression on social media can encourage opinion expression and action on important social issues online and offline.

These two forms of social media—consumptive and expressive—act in conjunction rather than individually or separately. When algorithms rely on both active and passive forms of social media use by its users (e.g., sharing or liking a link as the active form and clicking on a link as passive in Facebook) to design information feeds that users see, an individual's news environment becomes highly personalized (Hermida, Fletcher, Korell, & Logan, 2012). In the context of social media use related to information around important political and social issues, this means that an individual's online history and who they are friends with can play a role in how they interpret and perceive issues. Thus, the setting of social media is increasingly important for understanding public perceptions around the issue of climate change.

One important component of social media is the idea that featured information feeds will reinforce preferences of the user. People have a tendency to select and interpret information for all types of issues, including scientific ones, that supports their existing viewpoints (Yeo, Xenos, Brossard, & Scheufele, 2015). Scholars are concerned this process is exacerbated in the online media environment, in which users can personalize their information feeds by following, liking, and sharing information from other users who are like them, often referred to as an "echo chamber," in which individuals cluster among like-minded individuals (Sunstein, 2007). Recent evidence shows scientific news users engage in this behavior in social media sites like Facebook and YouTube (Bessi et al., 2016). Algorithms may also augment these processes by relying on users' preferences to portray future content, further reinforcing the content that users see to be like the content with which they mostly interact (Pariser, 2011). Research shows this may occur in search engine context

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for scientific issues, with the content of search engine results favoring keyword searches more over time (Ladwig, Anderson, Brossard, Scheufele, & Shaw, 2010), and it is likely to occur among social media feeds, as well.

Climate Change Public Opinion and Social Media

The Role of Personalization in Climate Change Opinion

Scholars recognize that climate change is an abstract topic for most, and public opinion about it forms more readily in the presence of making it psychologically closer to the individual. Information filtered through social media may be one of these personalizing and concretizing experiences that bring climate change closer to individuals.

Personalization can reduce the psychological distance between the person and climate change, which makes it easier for the person to engage with the issue (Spence, Poortinga, & Pidgeon, 2012). A related concept is experiential cognitive processing, a type of processing where individuals interpret information through affect and emotion, which makes it more relatable to them (Stanovich & West, 1998). This makes a distant and abstract issue like climate change more concrete in people's minds (Marx et al., 2007). When people perceive personal experiences with climate change, they are more likely to believe in the issue (Myers, Maibach, Roser-Renouf, Akerlof, & Leiserowitz, 2012). One important area where people concretize and make climate change a personal experience is the weather. Indeed, several studies have shown that when people associate weather-related events or temperatures with climate change, they are more likely to believe in climate change and perceive more risks from it (Borick & Rabe, 2014; Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Howe, 2013). Thus, one key area of scholarship in climate change public opinion is the idea that people form opinions about the issue when it is personalized or made more concrete for them.

Social media embeds news and information about social issues in a social context, which provides a personal context for individuals. Information is filtered through friends (Metzger et al., 2010), and sites such as Facebook provide information based at least partially on an individual's previous information habits (Pariser, 2011). In addition, social media use is often conducted in visual form, with half of social media users sharing or reposting news stories, images, or videos (M. Anderson & Caumont, 2014), and climate change is effectively communicated visually (van der Linden, Leiserowitz, Feinberg, & Maibach, 2014). In short, social media personalizes social issues in several ways, and is thus an appropriate lens through which to analyze climate change opinion.

Analyzing social media use raises the question: how much or how often do people actually discuss the topic of climate change over social media? Data available suggest 7% of Americans share information about global warming on Facebook or Twitter, and 6% of Americans have posted a comment online in response to a news story or blog about global warming (Leiserowitz, Maibach, Roser-Renouf, & Feinberg, 2013). These numbers are quite a bit lower than the frequency of discussion offline. About 35% of individuals in the United States report discussing climate change

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with family or friends at least occasionally (Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Rosenthal, 2015). Thus, individuals are not widely discussing climate change in social media spaces, but evidence suggests that a small percentage are.

Given that weather is a primary topic that personalizes the abstract issue of climate change for people, it is worth examining the frequency with which people discuss it over social media. Weather-related information consumption is moving online, even if television is still the primary source people use to seek out information about the weather (Harris Interactive, 2007; Lazo, Morss, & Demuth, 2009; Rosenstiel, Mitchell, Purcell, & Rainie, 2011). A quarter of the population in the United States checks the Internet at least once a day for weather information (Lazo, Morss, & Demuth, 2009). Of those who are already Internet users, weather news leads the topics they will read about online (Purcell, Rainie, Mitchell, Rosenstiel, & Olmstead, 2010). Evidence shows that people do talk to others about extreme weather events, and more than 30% of people in the United States who have experienced an extreme weather event talk about it over social media (Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Howe, 2013). More than 20% of Americans report sharing about a weather experience over social media, and nearly 2 in 10 (19%) have shared a photo of the event they have experienced (Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Howe, 2013). Fourteen percent of individuals report commenting about extreme weather events on blogs or news sites (Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Howe, 2013). The vast majority of people in the United States who have experienced one of these events have talked about it offline with someone else either in person (89%) or over the phone (84%) (Leiserowitz, Maibach, Roser-Renouf, Feinberg, & Howe, 2013). Thus, evidence shows that communication with others about extreme weather events is much higher offline than online. Nonetheless, people are discussing weather online. It is possible that as people make connections between weather and climate change, discussions about climate change online will increase. Indeed, one study that examined tweets in the United States provides evidence that people mention temperature anomalies and climate change together (Kirilenko, Molodtsova, & Stepchenkova, 2015).

Analyzing social media also provides a setting to examine what opinions and sentiment members of the public have about climate change. Several studies have assessed the sentiment of discussions about climate change or climate change events on different social media. We can better understand what people associate with climate change because of these discussions. One study that examined sentiment in Twitter posts showed that people talk negatively about the climate-related topics of natural disasters, oil drills, and climate bills but talk positively about climate rallies, green ideas, and a book release (Cody, Reagan, Mitchell, Dodds, & Danforth, 2015).[1] Both “climate change” and “global warming” tend to garner more negative tweets, such as global warming being catastrophic, than positive tweets, such as weather being more pleasant due to climate change (Lineman, Do, Kim, & Joo, 2015). The difference between negative and positive tweets, however, is greater for global warming than for climate change (Lineman et al., 2015). In all, there are a range of topics people associate with climate change in their social media discussions that tend to reflect a negative tone, although not exclusively a negative tone.

Other analyses of social media content examine how people frame the issue in their discussions. Again, some of these studies compare global warming to climate change. For instance, one analysis of tweets in the United Kingdom, United States, Canada, and Australia found the phrase *global*

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warming is typically associated with more "hoax" frames or the idea that the issue is a hoax than the phrase *climate change* is (Jang & Hart, 2015). A study of media coverage of the release of the Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report showed that discussions on Twitter tend to portray climate change as a scientific fact rather than frame it as uncertain (O'Neill, Williams, Kurz, Wiersma, & Boykoff, 2015). It is likely that such an event, however, draws more social media engagement from individuals who already believe in climate change. Other events are more likely to draw people with skeptical perspectives. "Climategate," or the release of emails from university climate scientists that climate skeptics claimed were evidence that scientists were manipulating data to support the idea of human-caused global warming, brought climate change to the forefront of new media discussions, particularly blogs and online reader comments from a U.K.–based tabloid (Koteyko, Jaspal, & Nerlich, 2013; Pew Research Center, 2009). In an analysis of reader comments posted on the website for the U.K. newspaper *Daily Mail*, the Climategate event gave light to comments that are skeptical of climate change, showing that social media spaces like commenting sections of newspapers provide an outlet for climate skepticism (Jaspal, Nerlich, & Koteyko, 2012). Analyses of YouTube comments of English-language videos around Climategate show people discussed it as an example of scientific fraud, where scientists are corrupt and dishonest, and as an example of a political scam, or a hoax that is part of a larger international governmental conspiracy (Porter & Hellsten, 2014). Commenters, however, did also discuss the other side of it, suggesting that media outlets were overhyping the event by promoting a climate skeptical perspective (Porter & Hellsten, 2014). The discrepancy seen in the two studies of two different events in how climate change is either accepted (release of the IPCC report) or denied (Climategate) in social media discussions indicates that these social media discussions are potentially happening among like-minded individuals rather than providing a setting for discussions that cross opinion boundaries. Indeed, evidence shows that Twitter users discussing climate change tend to exist within like-minded communities, with activists and skeptics of climate change remaining polarized (Williams, McMurray, Kurz, & Hugo Lambert, 2015). Porter and Hellsten (2014), however, claim that in the YouTube comments they analyzed, people did argue with those who hold different perspectives from their own, but that they did not change their opinions or positions. Furthermore, the arguments with those who hold differing opinions appear to solidify their existing opinions further (Porter & Hellsten, 2014). And discussions online tend to express strong perspectives. For instance, a 2013 study that analyzed posts on Twitter that mention climate change suggests that individuals tend to have strong opinions of belief in climate change (either believing in climate change or not believing in it, rather than having a neutral perspective on whether climate change is happening and human-caused) (Williams et al., 2015). Other evidence, however, shows perspectives in social content can be more neutral. Those who do comment on blogs and are skeptical of climate change tend to hold a range of views, with only around 10% of commenters of one blog hosted by a U.S.–based climate scientist claiming climate change is a scam or fraud and others taking a more modest view on climate change (Matthews, 2015). Thus, different sources of various social content encourage a range of perspectives and strength of viewpoints on belief in and concern over climate change, some of which are held more strongly than others.

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The Role of Elite Cues in Climate Change Public Opinion

We know that elite and mass media cues play a role in climate change perceptions. This essentially means that politicians, advocacy groups, and traditional mainstream media all play roles in how people perceive climate change (Brulle, Carmichael, & Jenkins, 2012). As a result, much attention has been paid to the relationship between mainstream media and public perspectives on the topic. For instance, Indian media portray a nationalistic attitude toward climate change with the idea that compliance with carbon emissions reductions will limit India's growth, and this frame plays into the larger public discourse on the topic (Billett, 2010). A study of Japanese media found a positive relationship between amount of coverage of climate change and public concern for it, and this was particularly the case in the short-term when it was a front-page story (Sampei & Aoyagi-Usui, 2009). In the social media realm, evidence shows non-elite social conversations online, or content generated by users not formally part of a mainstream media organization, that occur alongside mainstream media content can play a role in how people perceive issues in the news and the news itself (A. A. Anderson, Brossard, Scheufele, Xenos, & Ladwig, 2014; A. A. Anderson, Yeo, Brossard, Scheufele, & Xenos, 2016; Lee, 2012; Metzger et al., 2010; Walther & Jang, 2012). In other words, the tone of information sources outside of mainstream media (e.g., reader comments or social media posts) can shape attitudes. This suggests there is an interplay between the elite cue of mainstream media and the non-elite cue of social media discussions.

Scholars often analyze social media in relation to mainstream media. Discussions about climate change on Twitter spike around specific news stories or weather events, or the coverage of weather events (Kirilenko & Stepchenkova, 2014). Other events that tend to be covered in the mainstream media, such as the UN Climate Change Conference, encourage hashtag use on Twitter (Kirilenko & Stepchenkova, 2014). Scholars have also considered online searches in relation to mainstream media coverage. For instance, Lineman and colleagues (2015) analyzed how relative search volumes for global warming and climate change changed over time. They found that search volumes peaked during big media events, such as the release of the movie *Inconvenient Truth* and the selection of Al Gore and the International Governmental Panel on Climate Change for the Nobel Peace Prize (Lineman et al., 2015). Another study shows that from 2001 to 2009, people's interest in searching for nearly 20 environmental terms including *global warming* in Google declined, but that *climate change* searches increased (Mccallum & Bury, 2013). The study by Lineman and his colleagues (2015) that compared Google searches for *climate change* and *global warming* found a similar pattern for the two. Early searches were really predominated by the term *global warming* vs. *climate change*, but after 2009, amounts of searches for the two terms leveled off and became more similar to each other (Lineman et al., 2015). This indicates online interest in the topic is following broader mainstream media frames—which have started to focus more heavily on using the term *climate change* over *global warming*. It is likely that traditional media coverage encourages online media interest in the topic.

Social media is another (non-elite) cue that can be added to elite cues that impact climate change opinions. It is worth noting that social media cues can also be from elite sources, such as government organizations, and can be an effective avenue for building credibility for the organization and disseminating risk-based messages (e.g., Dalrymple, Young, & Tully, 2016). It is likely that climate change messages from elite organizations in social media can be effective means

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for shaping opinions alongside previously identified elite cues from more traditional sources (e.g., mainstream media), especially if individuals consider them to be highly credible. As Dalrymple and her colleagues (2016) noted, however, social media is not always the best venue for two-way communication between elite organizations and everyday users. Further empirical research testing effects of social media content on climate change opinion is warranted.

Climate Change Knowledge and Social Media

Within science communication, several scholars have examined the idea that Internet use reduces the knowledge gap, or the different rate that certain individuals are disadvantaged regarding learning about scientific issues due to aspects of their background, such as socioeconomic status (Cacciatore, Scheufele, & Corley, 2014). And in the literature on climate change public perceptions, in particular, there is evidence that Internet use is related to greater climate change knowledge. For instance, Zhao (2009) found that among Americans, Internet use, which includes social content such as conversations in chat rooms and on discussion boards, is associated with greater perceived knowledge of climate change. An extensive survey of Europeans found that Internet use is associated with greater levels of knowledge on topics such as carbon capture storage (Eurobarometer, 2011). Another study of Americans, however, that examined use of the Internet for news—and not necessarily social media—found a negative relationship between Internet use and climate change knowledge (Kahlor & Rosenthal, 2009). Thus, there is some evidence that the social aspect of online media use is a key factor in climate change knowledge. Additionally, recall of climate change information is higher when using visuals (van der Linden, Leiserowitz, Feinberg, & Maibach, 2014), a popular component of social media. Furthermore, there is other evidence that Internet use can be associated with lower levels of knowledge about climate change, depending upon other factors. For instance, when people perceive media to overplay the issue of climate change, Internet use is associated with lower levels of knowledge in a survey of Germans (Taddicken, 2013). Taddicken's (2013) study, however, found evidence for only this moderated relationship and did not find a direct relationship between Internet use and climate change knowledge. Thus, Internet use is not always related to increased knowledge of climate change, but several studies show that it generally is. While most of these studies discuss Internet use generally rather than social media use specifically, Zhao's (2009) study suggests that social content is associated with higher knowledge.

Research indicates social media contributes to the specialized knowledge of climate change and the development of knowledge communities around climate change. Indeed, Twitter streams around major climate change events such as marches during the 2009 Fifteenth Session of the Conference of the Parties to the United Nations Framework Convention on Climate Change (COP 15) that were identified through hashtags have been referred to as "long-running epistemic communities" (Segerberg & Bennett, 2011). In an analysis of Twitter posts about climate change, skeptics of climate change tended toward specific hashtags like #agw (which stands for "anthropogenic global warming") or #climaterealists rather than more general hashtags such as #climatechange or #globalwarming (Williams, McMurray, Kurz, & Hugo Lambert, 2015). This supports the idea that knowledge communities for specific perspectives on global warming exist in

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the social media environment. Thus, elements of expressive social media use like hashtags encourage the development of spaces for those communities. Furthermore, a survey of scientists shows that readership of scientific blogs is related to greater awareness of specific instances of political conflict around climate change (Nisbet & Markowitz, 2015). This points to the idea that blogs contribute to specialized knowledge.

There is also evidence that Internet use can provide space for cognitive processing that is more reflective and based on reason. While this advanced cognitive processing is not a direct measure of knowledge, it is a component of more critical thinking about the issue. Some studies provide evidence that online comments about climate change can be deliberative, meaning that they consider a variety of perspectives on the topic rather than displaying a narrow perspective (Collins & Nerlich, 2014; Graham & Wright, 2015). This provides support for the idea that consumption in the online social context can provide a space for reasoned reflection or cognitive processing of the issue of climate change. Expressive social media use also provides space for contributing to a deliberative community. For instance, an online tool called Climate Feedback encourages scientists to add annotations, such as further evidence or support for or disagreement with a published perspective, to stories in major news outlets such as the *New York Times, Wall Street Journal*, or CNN (Readfearn, 2015). Several climate scientists using the tool all agreed a *Daily Telegraph* story on Earth heading for a mini–ice age to be low or very low in credibility (Readfearn, 2015). The open science movement also suggests Internet-based platforms can provide more transparency to the scientific process (Saraga, 2016), potentially providing space for more knowledgeable debate about the often controversial issue of climate change. While spaces for deliberation are not directly related to knowledge, reflective cognitive processing that can occur in deliberative spaces indicates people are thinking more deeply about a topic. Such cognitive processing is a positive outcome that can relate to greater knowledge.

A related concept to reasoned reflection is that Internet use can encourage habits or behaviors related to knowledge, such as information seeking. For instance, Zhao (2009) found that Internet use is related to greater likelihood to seek out more information on the topic, as mediated by perceived knowledge. Indeed, having an attitude that seeking information about global warming is valuable, beneficial, or good and understanding information found in media is associated with higher knowledge of climate change (Kahlor & Rosenthal, 2009).

In short, plenty of evidence exists that Internet use is associated with greater knowledge about climate change—both direct or perceived knowledge and other indicators of knowledge, such as reasoned reflection and specialized knowledge communities. Much of this research examines general Internet use or news use on the Internet, and it remains to be seen whether the online social content, in particular, performs differently when measuring knowledge levels. Further research examining relationships between social media use and climate change knowledge is needed.

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Climate Change Behavior and Social Media

Generally, relatively few studies have examined the relationship between Internet use or social media use and climate change activism or engagement, even though several studies on a range of political issues have shown a relationship between social media use and political action (Boulianne, 2015). For the issue of climate change, one study showed that frequent usage of online versions of traditional media outlets was positively associated with intentions to adopt more political behaviors related to climate change, such as becoming active in promoting policy change (Arlt, Hoppe, & Wolling, 2011). This same study, however, did not find any significant relationship between online media use and adopting a different lifestyle that would help mitigate climate change (e.g., driving less or purchasing energy-saving light bulbs) (Arlt et al., 2011). Some studies take the approach of examining how online activities support offline activism related to climate change. Internet-based advocacy organizations for climate change (e.g., 350.org), which at least sometimes utilize social media to communicate with or to members of the public, are successful in mobilizing offline publics and therefore influencing policy debates around issues such as the Keystone XL pipeline (Hestres, 2014). Some suggest such online activities should complement offline activism and mobilization and not replace it (Nisbet & Kotcher, 2009). Furthermore, individuals use components of Twitter such as hashtags, retweets, and replies to provide information and organize logistics of offline activities, like climate change marches around major events like the COP 15 conference held in Denmark in 2009 (Segerberg & Bennett, 2011). Another study found a relationship between using Facebook for engaging with energy issues (e.g., liking energy pages and sharing energy-related news content) and perceiving oneself as an opinion leader and feeling a sense of efficacy for political actions related to climate change (Vraga, Anderson, Kotcher, & Maibach, 2015). In all, there is some evidence for a relationship between social media use in particular—and not just online media use—and taking action on climate change.

Studies also examine how online activities spur activism that resides primarily within the online realm. For instance, Katz-Kimchi and Manosevitch (2015) analyzed an online activist campaign by Greenpeace that protested Facebook's use of coal as a primary energy source using communication over Facebook itself. They claimed that this tactic was productive in mobilizing a broader set of activists for the campaign outside of the professional activist realm and without the need to rely on mainstream media sources to share information (Katz-Kimchi & Manosevitch, 2015). Another study identified collective action frames, such as information sharing and sharing evidence, in YouTube comments about Climategate, a mediated event without accompanying offline activities (Porter & Hellsten, 2014). Thus, social media discussions by both professional environmental organizations and by regular Internet users encourage different forms of online action on the issue of climate change. This perspective does not acknowledge, however, that online activism of the information sharing nature may not necessarily account for much tangible change on the issue—otherwise known as slacktivism (McCafferty, 2011). Thus, future research should further assess the meaning of these online forms of action when they stand alone.

There is some evidence that discussions over social media are not of the nature in which a range of individuals have a platform for communicating about climate change. For instance, some have argued social media is useful for activating one's own opinion group but not necessarily for talking

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to individuals who do not share one's opinion. One study of climate change discussions over Twitter provides evidence that people tend to talk to others who are like-minded, with a minority of users talking to people who share diverse views (Williams, McMurray, Kurz, & Hugo Lambert, 2015). Others argue that opinion leaders on social media for climate change are few in number and highly concentrated among elite journalists, organizations, and celebrities—as evidenced by an analysis of Twitter discussions in which relatively few profiles dominate the discussion (Kirilenko & Stepchenkova, 2014). Both of these studies suggest the potential for activism via social media may be limited to mobilizing among an already-committed group of individuals.

Opinion leaders are those who are "likely to influence other persons in their immediate environment" (Katz & Lazarsfeld, 1955, p. 3), and are an important component of influencing a variety of actions on climate change, including consumer behaviors and political participation among others (Nisbet & Kotcher, 2009), and are thus a significant part of any discussion on climate change behavior. Social media is a space where opinion leaders can share information about how to get involved in an issue such as climate change and expose peers to information they otherwise would not see (Nisbet, Markowitz, & Kotcher, 2012). Indeed, empirical research shows that opinion leaders on social media such as Twitter are known for being more motivated to use the service to seek information, mobilize others, and express their opinion on it (Park, 2013). In addition, people trust media stories they get on social media from friends who they perceive to be opinion leaders (Turcotte, York, Irving, Scholl, & Pingree, 2015). Internet communication, including social media, is also a useful tool for environmental nongovernmental organizations trying to communicate, coordinate, and mobilize climate activists (Castells, 2009; Hestres, 2015).

In sum, there is initial evidence that social media is productive in encouraging more environmentally friendly behaviors that will mitigate climate change and in sparking activism around the issue of climate change. Scholars are optimistic that new media provide opportunities to increase information sharing, participation, and engagement with climate change (O'Neill & Boykoff, 2011). The future direct empirical research will help solidify these early perspectives on the topic.

Conclusion

There is a wide range of possible roles social media can play in encouraging different attitudes and behaviors around climate change, and there are many unanswered questions in these areas of research. First, parsing out the effects of different content-related components of climate change communication in social media is crucial for better understanding the role of social media in public perceptions about climate change. This review suggests a negative tone in the sentiment of discussions on climate change, but it is difficult to tell the nature of that negativity. Scholars suggest that language and iconic visuals that are alarmist in nature or rely on fear appeals may raise concerns but also disengage audiences (Nerlich, Koteyko, & Brown, 2010; O'Neill & Nicholson-Cole, 2009). While there is reason to be optimistic about the ability of social media to positively influence opinion, knowledge, and behavior around climate change, some caution that social media use may simply encourage more reinforcement of existing perceptions of climate change rather than reaching new individuals or changing opinions (Nisbet, Markowitz, & Kotcher, 2012). Indeed,

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research in other risk contexts provides evidence that online social content in news comments can polarize existing risk perceptions (A. A. Anderson, Brossard, Scheufele, Xenos, & Ladwig, 2014), suggesting that social media may exacerbate existing divisions in society. While this could be a boon for mobilizing those already concerned about climate change, care should be given to considering the extent to which social media provides a skeptical perspective of climate change and the effect this may have on people who have not yet made up their minds about the issue. In addition, it could be that certain aspects of tone of social content (e.g., hopeful emotion in a YouTube comment) positively engage others who have not thought much about the topic while other tones (e.g., sarcasm in a news comment) turn people off to any further engagement with the issue. Digging deeper into these sentiment analyses to better understand the negativity in them can inform future research on uncovering effects of social media on climate change perceptions and behaviors. The second area of research on social media that climate change communication researchers should focus on is source-related questions. Where do attitudes about different levels of source (e.g., the linked content or the peer sharing the link) on social media come from, and how do those different sources affect public perceptions of climate change? A range of information sources are involved in social media communication, and answering some of these source-related questions is crucial to understanding the role of social media in climate change perceptions. A third essential area of research is understanding how visual communication of social media can influence various public perceptions. Social media is an obvious space for the kinds of local and relatable language and visuals that can be effective in activating individuals on climate change (Nerlich et al., 2010; O'Neill & Nicholson-Cole, 2009). For instance, weathercasters and news outlets can discuss and encourage photo and video social media posts about regional extreme weather events from users. Social media is also a natural space to break down complex scientific information visually and graphically and reach large audiences through sharing and discussion.

In conclusion, studies of climate change–related online social content ranging from Twitter discussions to news comments to online searches provide evidence for relationships between social media use and climate change opinion, knowledge, and behavior. Given that research on the effects of online communication generally—and social media in particular—on climate change perceptions is minimal (Schäfer, 2012), further research will more clearly identify common patterns of these relationships. Early evidence suggests several positive impacts, with social media encouraging greater knowledge of climate change, mobilization of climate change activists, space for discussing the issue with others, and online discussions that frame climate change as a negative for society. Social media, however, does provide space for framing climate change skeptically and activating those with a skeptical perspective of climate change. Further examining and delineating these positive and negative aspects of social media for climate change perceptions is warranted.

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Notes

1. Many of the studies that analyze sentiment over social media do not explicitly specify whether the posts being analyzed are solely English-language posts, although that is a fair assumption considering that the analyses describe patterns and findings in English. Mention of the origin or language of the social media being analyzed is given here when specifically mentioned in the study.

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